

Seizure detection using scalp-EEG

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Summary

Scalp electroencephalography (EEG)-based seizure-detection algorithms applied in a clinical setting should detect a broad range of different seizures with high sensitivity and selectivity and should be easy to use with identical parameter settings for all patients. Available algorithms provide sensitivities between 75% and 90%. EEG seizure patterns with short duration, low amplitude, circumscribed focal activity, high frequency, and unusual morphology as well as EEG seizure patterns obscured by artifacts are generally difficult to detect. Therefore, detection algorithms generally perform worse on seizures of extratemporal origin as compared to those of temporal lobe origin. Specificity (false-positive alarms) varies between 0.1 and 5 per hour. Low false-positive alarm rates are of critical importance for acceptance of algorithms in a clinical setting. Reasons for false-positive alarms include physiological and pathological interictal EEG activities as well as various artifacts. To achieve a stable, reproducible performance (especially concerning specificity), algorithms need to be tested and validated on a large amount of EEG data comprising a complete temporal assessment of all interictal EEG. Patient-specific algorithms can further improve sensitivity and specificity but need parameter adjustments and training for individual patients. Seizure alarm systems need to provide on-line calculation with short detection delays in the order of few seconds. Scalp-EEG-based seizure detection systems can be helpful in an everyday clinical setting in the epilepsy monitoring unit, but at the current stage cannot replace continuous supervision of patients and complete visual review of the acquired data by specially trained personnel. In an outpatient setting, application of scalp-EEG-based seizure-detection systems is limited because patients won't tolerate wearing widespread EEG electrode arrays for long periods in everyday life. Recently developed subcutaneous EEG electrodes may offer a solution in this respect.

KEYWORDS

detection delay, epilepsy monitoring unit, false alarm rate, sensitivity

1 | INTRODUCTION

Seizure detection using scalp electroencephalography (EEG) can theoretically be applied either during inpatient long-term video-EEG monitoring (VEM) in the epilepsy monitoring unit (EMU) or in an outpatient setting. During

VEM in the EMU, potential indications include improvement of patient safety, more efficient data analysis, objective quantification of seizures, and automatic testing during seizures. In an outpatient setting, scalp-EEG-based seizure detection could facilitate objective seizure documentation and quantification, as well as seizure alerting or warning

systems to improve patient safety and eventually trigger on-demand therapeutic interventions.

2 | INDICATIONS FOR SCALP-EEG-BASED SEIZURE DETECTION

Patient safety with the goal of reducing the risks and consequences of seizures including seizure-related injuries, cardiac arrhythmias, and, in rare cases, sudden unexpected death in epilepsy (SUDEP) is of critical importance during VEM.¹⁻⁴ Although there is consensus that patient safety can be optimized by continuous surveillance through trained personnel,^{4,5} according to several surveys, constant observation is available in only 56%-80% of EMUs.^{2-4,6} Therefore, automatic on-line EEG seizure detection and warning systems could significantly improve patient safety. However, such systems are used in only 15%-19% of EMUs.^{3,6}

Automatic scalp-EEG-based seizure detection could also help optimize the time-consuming review and analysis of the large amount of data acquired during VEM and therefore significantly reduce the workload in EMUs.⁷⁻⁹ Although agreement on the annotation of EEG seizures during VEM between human experts is high, with an average any-overlap sensitivity of 92% and false-positive per hour rate of 0.117 applying any-overlap comparisons (ie, whether there was any overlap between experts of detection during a period annotated as a seizure), high seizure rates (many per hour), short seizure durations (<10 seconds), and long seizure durations (~10 minutes) with ambiguous offsets can complicate analysis and result in poor correlations between electroencephalographers.¹⁰ Automatic seizure detection systems could help to objectively quantify seizures during VEM.

Clinical seizure semiology provides important information on the seizure-onset zone and pathways of seizure propagation. However, the full information contained in clinical seizure semiology can be obtained only if clinical behavior during and after seizures is assessed and patients are interactively tested in a systematic way.^{11,12} According to a recent study, behavioral testing with questions or commands was performed in only 50% of seizures ictally, in 73% of seizures postictally, and in 80% of seizures either ictally or postictally.¹³ Therefore, a series of video-recorded behavioral tasks automatically triggered to play in the patient's room by computerized seizure detection (Automatic Responsiveness Testing in Epilepsy [ARTiE]) has been developed, which could improve periictal testing.¹³

In an outpatient setting, one of the major problems in clinical epileptology is that seizure documentation and counting by patients is extremely unreliable. In a VEM study, patients failed to document 55.5% of all recorded seizures.¹⁴ In a seminal study using a long-term, implanted

Key Points

- Scalp-EEG-based seizure detection algorithms provide sensitivities between 75% and 90%
- Detection algorithms generally perform better on seizures of temporal lobe origin as compared to those of extratemporal origin
- Specificity (false-positive alarms) varies between 0.1 and 5 per hour and is of critical importance for clinical acceptance of algorithms
- Seizure alarm systems need to provide on-line calculation with short detection delays in the order of few seconds
- In an outpatient setting, use of scalp-EEG-based seizure-detection is limited because patients won't tolerate wearing EEG electrode arrays in everyday life

seizure advisory system in patients with drug-resistant epilepsy, disparities in seizure frequency between patients' seizure diaries and EEG documented seizures were often substantial, with most patients underestimating the frequency of their seizures. This relation varied highly from month to month, preventing the hypothetical application of a correction factor.¹⁵ Therefore, objective measures of seizure documentation and counting are needed, also in order to obtain efficacy measures of therapeutic interventions.¹⁶ Automatic seizure detection could also be used as seizure alerting or warning systems to increase patients' quality of life by lowering anxiety due to the unpredictability of seizures and their consequences. Furthermore, automatic seizure detection devices could help prevent seizure-related injuries and eventually SUDEP.^{17,18} Finally, a seizure detection system could pave the way to on-demand therapies, such as acute release of anticonvulsants or electrical stimulation, to interrupt seizures or prevent seizures from occurring at all.¹⁶

3 | ALGORITHMS FOR SCALP-EEG-BASED SEIZURE DETECTION

Algorithms using single versus multiple EEG channels can be distinguished. Notably, some of the proposed algorithms are based essentially on single-channel analyses using bipolar montages.¹⁹⁻²² However, virtually all algorithms used in a clinical setting used multiple channels for detection. When using multiple EEG channels, the montage can be of critical importance for performance of the algorithm.²³ Furthermore, both for computational reasons and for patient comfort (especially in outpatient settings), a selection and

reduction of the number of electrodes is desirable.^{24–26} Use of only 8 frontal and temporal electrodes yielded an average detection sensitivity of 79%; and use of only 7 temporal, parietal, and occipital electrodes an average detection sensitivity of 68%, as compared to a sensitivity of 84% for all 21 channels of the International 10-20-system.²⁵

After data acquisition, all algorithms apply some kind of artifact rejection. Thereafter, detection of EEG-seizure patterns is usually based on characteristic electrographic changes during seizures with respect to frequency, amplitude, and/or rhythmicity. Linear and nonlinear time-frequency signal analyses techniques with numerous features have been introduced and will be summarized in the following sections.^{22,27,28}

4 | PERFORMANCE MEASURES

The gold standard for performance evaluation of seizure-detection algorithms remains visual annotation of seizures by EEG experts. Although agreement between EEG experts is usually high, discrepancies can occur in the event of high seizure rates and seizures of short and long durations, which needs to be considered during evaluation of computer-based seizure-detection algorithms.¹⁰ Ideally, several blinded EEG reviewers should obtain consensus on visual seizure annotations.^{8,29}

The following measures are usually applied to determine the performance of a seizure-detection algorithm²⁷:

- True-positives (TPs), that is, EEG seizures both annotated by human experts and detected by the algorithm;
- False-negatives (FNs), that is, EEG seizures annotated by human experts, but not detected by the algorithm;
- False-positives (FPs), that is, events erroneously identified as seizures by the algorithm;
- Detection sensitivity referring to the ratio of TP/(TP + FN);
- Specificity defined as the number of false-positive alarm rate per hour (FAR). In the following all FAR values mentioned refer to false-positive alarm rate per hour.

If an algorithm is designed to work as an alarm device, on-line calculation and short-detection delays are of critical importance. Detection delay is defined as the time interval between visually identified EEG seizure onset and the time when the alarm is set by the algorithm. Detection delays need to be in the range of a few seconds for an alarm device.^{30,31}

An algorithm should detect a broad range of seizures in patients with different epilepsy syndromes and seizure-onset zones with high sensitivity and high selectivity without the necessity to have a priori knowledge about the

patient's individual electrographic seizure patterns. Furthermore, an algorithm should be easy and ready to use with identical parameter settings for all patients. Analysis of large EEG datasets should be fast and robust. All EEG data acquired during VEM should be analyzed, including all kinds of artifacts, different neurophysiological states, as well as nonictal physiological and pathological EEG patterns.²⁷

On the other hand, the performance of algorithms can significantly be improved by parameter adjustments for individual patients, providing patient-specific detection.^{32–34} However, such patient-specific detection algorithms need specific training and interactive, sometimes complex, individual parameter adjustments.

5 | NON-PATIENT-SPECIFIC ALGORITHMS TESTED IN CLINICAL SETTINGS

Jean Gotman was the first to introduce scalp-EEG-based seizure detection in a clinical setting and published a series of seminal papers on this topic (Table 1).^{35–37} His first algorithm was based on decomposition of the EEG into half waves and detection of paroxysmal rhythmic activity with a frequency between 3 and 20 Hz. The algorithm was tested in 24 scalp-EEG recordings from 16 patients with an average recording duration of 12.4 hours. The system was capable of on-line detection of numerous seizure types. False detections due to nonepileptiform rhythmic EEG bursts and artifacts were quite frequent.³⁵ In 1990, improvements of the algorithm were reported, taking into account a larger temporal context and thus improving specificity. The algorithm was evaluated on 241 scalp-EEG recordings from 44 patients including 179 seizures, totaling 4362 hours of scalp EEG. Detection sensitivity was 73.2% with a FAR of 0.84/h. Of note, in 38.5% of the seizures, the patient alarm was not pressed, but the program detected the seizure.³⁷

Using Gotman's algorithm, Pauri et al⁷ investigated 461 hours of EEG of 12 patients with medically refractory focal epilepsy containing 253 EEG focal seizures. By testing different parameter settings, they obtained an optimum detection sensitivity of 81.4% and a FAR of 5.38/h. There was no relationship between EEG seizure patterns and performance of the algorithm. Detection was facilitated using sphenoidal electrodes, longer seizure duration, and shorter time from seizure onset to its full development.

Gabor et al³⁸ developed a self-organizing neural-network algorithm (CNET). The algorithm was tested in 200 records from 65 patients (4553.8 hours of recording containing 181 seizures).³⁹ CNET detected 92.8% of seizures with a FAR of $1.35/h \pm 1.35/h$.

TABLE 1 Non-patient-specific, scalp-EEG-based seizure-detection algorithms tested in clinical settings

Reference	Algorithm used	EEG sample (h)	Patients	Seizures	Sensitivity (%)	Specificity FAR (per h)	Detection delay (s)
Gotman ³⁷	Decomposition of EEG into half waves → measuring rhythmicity of signal	4362	44	179	73.2	0.84	n.a.
Pauri et al ⁷	Decomposition of EEG into half waves → measuring rhythmicity of signal	461	12	253	81.4	5.38	n.a.
Gabor et al ^{38a}	Neural networks—CNET	528	22	62	90.3	0.71	n.a.
Gabor ^{39b}	Neural networks—CNET	4554	65	181	92.8	1.35	n.a.
Wilson et al ⁴⁰	Matching Pursuit, small neural networks, connected-object hierarchical clustering—Reveal	1049	426	672	76.0	0.11	n.a.
Saab and Gotman ³⁰	Wavelet transforms → probabilistic Bayesian detection	360	16	69	76.0	0.34	10.0
Kuhlmann et al ⁴¹	Wavelet transforms → probabilistic Bayesian detection	525	21	88	81.0	0.60	16.9
Meier et al ³¹	Wavelet transforms, mean sliding variance, mean cross-correlation, Savitzky-Golay filter, zero-crossings → support vector machine	1403	57	91	>96.0	<0.5	2.0
Kelly et al ⁸	Pattern-match regularity statistics, local maximum frequency, amplitude variation—IdentEvent	1200	55	146	79.5	0.08	n.a.
Zandi et al ²⁸	Normalized wavelet-based index—combined seizure index (CSI)	236	26	79	91.0	0.33	7.0
Hopfengärtner et al ²³	Power spectral analytical techniques—Short time Fourier transform	3248	19	148	90.9	0.29	19
Hopfengärtner et al ²⁷	Power spectral analytical techniques—Short time Fourier transform	25 278	159	794	87.3	0.22	n.a.
Hartmann et al ⁴²	Time and frequency domain analysis—Epileptiform wave sequence (EWS) analysis and periodic waveform analysis (PWA), EpiScan	4300	48	186	83.0	0.3	n.a.
Fürbass et al ⁴³	Time and frequency domain analysis—Epileptiform wave sequence (EWS) analysis and periodic waveform analysis (PWA), EpiScan	22 000	275	623	73.0	0.30	n.a.
Fürbass et al ^{9b}	Time and frequency domain analysis—Epileptiform wave sequence (EWS) analysis and periodic waveform analysis (PWA), EpiScan	15 684	205	526	81.0	0.29	n.a.
Fürbass et al ^{9a}	Time and frequency domain analysis—Epileptiform wave sequence (EWS) analysis and periodic waveform analysis (PWA), EpiScan	25 567	310	113	75.0	0.30	n.a.

FAR, false-positive alarms per h.

^aDevelopment dataset.^bProspective dataset.

Wilson et al⁴⁰ introduced the Reveal algorithm utilizing 3 novel methods for seizure detection, namely Matching Pursuit, small neural network-rules, and a new connected-object hierarchical clustering algorithm. The system was tested on a total of 1049 hours of EEG from 426 epilepsy patients containing 672 seizures and showed a sensitivity of 76% with a FAR of 0.11/h.

Saab and Gotman³⁰ determined seizure probability of a section of EEG based on the values of 3 features, namely, relative average amplitude (RAA), relative scale energy (RSE), and coefficient of variation of amplitude (CVA). Analysis performed on 360 hours of scalp EEG, including 69 seizures in 16 patients showed a sensitivity of 77.9%, a FAR of 0.86/h, and a median detection delay of 9.8 seconds. In patients having unacceptable high FAR, a tuning mechanism provided significant reduction of FAR with minimal loss of detection sensitivity and detection delay (sensitivity 76.0%, FAR 0.34/h, median detection delay 10 seconds).

Kuhlmann et al⁴¹ used a similar probability-based framework and evaluated the performance of different combinations of 9 wavelet-based features including those used by Saab and Gotman.³⁰ The method was evaluated on 525 hours of EEG in 21 patients including 88 seizures. A detector based on an alternative combination of features achieved a sensitivity of 81%, a FAR of 0.60/h, and a median detection delay of 16.9 seconds.

Meier et al³¹ took into account different seizure morphologies (including rhythmic activities in the alpha-, beta-, theta-, and delta bands; amplitude depression; and runs of polyspikes). Their algorithm was based on 7 different features (including power of continuous wavelet transforms at specific scales, mean sliding variance, mean cross-correlation, the Savitzky-Golay filter, and zero-crossings) and used a support vector machine for seizure classification. EEG recordings from 57 patients (including 91 seizures [37 focal, 54 secondarily generalized]) with a duration of approximately 43 hours and an additional 1360 hours of seizure-free EEG data were analyzed. Average correct detection rate was >96%, mean FAR <0.5/h, for specific ictal morphologies <0.25/h, median detection latency of 2.0 seconds. The authors concluded that taking seizure morphology into account played a crucial role in increasing detection performance of the system and that their procedure facilitated automatic categorization of prevalent seizure morphologies without the necessity of adapting the system to specific patients.

Kelly et al⁸ developed an off-line, automated scalp-EEG-based seizure detection system (IdentEvent) processing multichannel scalp-EEG signals and translating them into 3 main EEG descriptors, namely pattern-match regularity statistic (PMRS as a measure of signal complexity), local maximum frequency (LMF), and amplitude variation

(AV; the latter 2 used mainly for artifact rejection). The system was evaluated on 436 EEG studies obtained from 55 subjects (~1200 hours with 146 seizures). All EEG recordings were reviewed by 3 independent epileptologists; the presence or absence of seizures in each epoch was determined by majority rule. Detection sensitivity of IdentEvent was 79.5% with a very low FAR of 0.08/h.

Zandi et al²⁸ proposed an algorithm relying on a normalized wavelet-based index, named the combined seizure index (CSI). CSI is sensitive to both rhythmicity and relative energy of an EEG epoch as well as to the consistency of EEG patterns among different channels increasing significantly as seizures occur. The method was evaluated on 236 hours of scalp EEG from 26 patients including 79 focal seizures. Sensitivity was 91%, FAR 0.33/h, and median detection latency 7 seconds. Furthermore, average CSI at seizure onset was significantly larger on the side of the focus in patients with temporal lobe epilepsy (TLE), indicating lateralizing capability of this index in TLE.

Hopfengärtner et al²³ calculated the integrated power in 2 frequency bands (ranging from 3 to 12 Hz and from 12 to 18 Hz) for 3 different multichannel seizure detection montages with identical parameters for all montages and all patients. A total of 3248 hours of scalp EEG from 19 patients containing 148 seizures were analyzed. The highest sensitivity of 90.9% was obtained for a seizure detection montage including basal temporal electrodes referenced against the average of Fz-Cz-Pz in the frequency band from 3 to 12 Hz with a FAR of 0.29/h. Median detection delay ranged between 10 and 44 seconds depending strongly on individual seizure patterns (overall median 19.0 seconds). The authors concluded that the chosen seizure detection montage had a significant influence both on sensitivity and selectivity, and that averaged sensitivity values were higher for the frequency band ranging from 3 to 12 Hz. Hopfengärtner et al²⁷ improved this algorithm by an adaptive thresholding technique and calculated the integrated power in the frequency band from 2.5 to 12 Hz for the seizure-detection montage mentioned above including basal temporal electrodes referenced against the average of Fz-Cz-Pz. A total of 25 278 hours of EEG data from 159 patients (117 temporal lobe epilepsies [TLEs]; 35 extratemporal lobe epilepsies [ETLE]; 7 others) containing 794 seizures were analyzed. Sensitivity was 87.3% and FAR was 0.22/h. For TLE patients (18 996 hours of EEG including 589 seizures), sensitivity was 89.9% and FAR 0.19/h; for ETLE patients (5192 hours of EEG including 172 seizures) sensitivity was 77.4% and FAR 0.25/h. The reduced seizure detection montage using only the 10-20 electrodes without temporal basal electrodes provided a sensitivity of only 78.8% and a FAR of 0.11/h.

Hartmann et al⁴² developed an online seizure-detection algorithm (EpiScan) based on a periodic waveform analysis

detecting rhythmic EEG patterns and an adaptation module automatically adjusting the algorithm to patient-specific EEG properties. The algorithm was evaluated on 4300 hours of unselected EEG recordings from 48 patients with TLE containing 186 seizures. Mean sensitivity was 83% and FAR 0.3/h. For 66% of the patients the algorithm detected 100% of seizures. Subsequently, Fûrbass et al⁴³ introduced a time-domain method for detection of seizure patterns called epileptiform wave sequence (EWS) analysis with focus on irregular and distorted rhythmic activity. The method scans the EEG for sequences of similar epileptiform discharges and uses a combination of duration and similarity measures for seizure detection. The method was tested on 22 000 hours of unselected and uncut scalp EEG recordings from 275 patients including 623 seizures. Used in combination with the EpiScan algorithm, sensitivity could be increased from 70% to 73% while FAR could be reduced from 0.33/h to 0.30/h. Fûrbass et al⁹ conducted a prospective multicenter study using EpiScan. A total of 15 684 hours of EEG recording from 205 patients (including 94 patients with 526 seizures) were evaluated. A mean sensitivity of 81% and a FAR 0.29/h could be achieved. Of particular notice, EpiScan detected 16 seizures that were previously undetected during routine visual analysis (3% of all seizures). For the development dataset containing 25 567 hours of EEG from 310 patients (including 124 patients with 1113 seizures), a mean sensitivity of 75% and a FAR of 0.3/h could be obtained. For this dataset, sensitivity for TLE (83%) was better than for extra-TLE (ETLE) (64%).

Commercially available seizure-detection algorithms are summarized in Table 2.

6 | PATIENT-SPECIFIC ALGORITHMS TESTED IN CLINICAL SETTINGS

Qu and Gotman³² addressed the problem of frequent false alarms in particular patients, often due to repeated paroxysmal, but nonepileptiform patterns (Table 3). They introduced a program that learned about false detections occurring in the first day of a VEM session and attempted to eliminate similar patterns occurring during the remainder

of the session. They evaluated their method in 20 patients having particularly high false-detection rates. Seventy EEG sessions from 10 patients with scalp electrodes covering a total of 1071 hours were used for evaluation. FAR were reduced from 2.90/h to 1.40/h. Qu and Gotman^{44,45} further refined patient-specific algorithms, relying on a sample seizure and a set of nonseizure EEG segments, recorded evenly over 24 hours, which were used to train a modified nearest-neighbor classifier. The method was evaluated in 12 patients with a total of 47 seizures. The system reached a detection rate of 100% with an average detection delay of 9.35 seconds. The average FAR was only 0.02/h.

Shoeb et al⁴⁶ presented a patient-specific method using a wavelet decomposition to construct a feature vector that captured the morphology and spatial distribution of an electroencephalographic epoch, and then determined whether that vector was representative of a patient's seizure or non-seizure EEG using the support vector machine classification algorithm. The method was tested on 60 hours of scalp EEG from 36 pediatric patients including 139 seizures. Sensitivity was 94%, FAR 0.25/h, and detection latency 8.0 ± 3.2 seconds.

Khamis et al³³ investigated a novel application of autoregression (AR) spectral techniques in 12 patients with left TLE. The Burg maximum entropy AR method was applied to signals from 4 electrodes near the epileptic focus for each patient, and the AR spectra were parameterized based on scalp-EEG features described by a neurologist, thus mimicking clinical seizure identification. The parameters measured spectral peak power, sharpness, and location in a delta/low theta frequency range. An optimized nonlinear seizure-detection index, which accounted for spatial and temporal persistence of behavior, was then calculated. Performance was optimized using recordings from 2 patients (315 hours, 18 seizures). For the remaining 10 patients (1624 hours, 83 seizures), results were presented as a receiver-operating characteristic graph, yielding a sensitivity of 91.57% and a FAR of 0.273/h.

Minasyan et al³⁴ proposed a patient-specific method based on man. The algorithm was trained both for the period immediately preceding seizure onset and for the period immediately after seizure onset and tested for continuous scalp-EEG recordings (duration 15-62 hours, median 25 hours) from 25 patients including a total of 86 seizures. In

TABLE 2 Commercially available seizure-detection algorithms

Name of algorithm	Reference	Company	Internet address
Persyst-Reveal	Wilson et al ⁴⁰	Persyst	www.persyst.com
IdentEvent	Kelly et al ⁸	Optima Neuroscience	www.optimaneuro.com
BESA	Hopfengärtner et al, ²³ Hopfengärtner et al ²⁷	BESA	www.besa.de
EpiScan	Hartmann et al, ⁴² Fûrbass et al, ⁴³ Fûrbass et al ⁹	Encevis	www.encevis.com

TABLE 3 Patient-specific, scalp-EEG–based seizure-detection algorithms tested in clinical settings

Reference	Algorithm used	EEG sample (h)	Patients	Seizures	Sensitivity (%)	Specificity FAR (per h)	Detection delay (s)
Qu and Gotman ³²	Time and frequency domain features → recognize patient-specific false detections based on similarities between EEG patterns	1071	10	n.a.	n.a.	1.40	n.a.
Qu and Gotman ⁴⁴	Time and frequency domain features → modified nearest-neighbor classifier	29.7	12	35	100	0.03	9.5
Qu and Gotman ⁴⁵	Time and frequency domain features → modified nearest-neighbor classifier	n.a.	12	47	100	0.02	9.35
Shoeb et al ⁴⁶	Wavelet decomposition → support vector machine	60	36	139	94	0.25	8.0
Khamis et al ³³	Burg maximum entropy autoregression (AR) method → optimized nonlinear seizure-detection index	1624	10	83	91.6	0.27	n.a.
Minasyan et al ³⁴	Recurrent neural networks using a variety of predefined input features	625	25	86	100	0.02	4.0

FAR, false-positive alarms per h.

14 of the 25 patients, all seizures could be detected before seizure onset with a median pre-onset time of 51 seconds and a FAR 0.06/h. Post-onset detection had 100% sensitivity, a FAR 0.023/h, and median delay of 4 seconds after onset.

7 | CONCLUSIONS

Scalp-EEG–based seizure detection algorithms provide sensitivities between 75% and 90%.⁴⁷ Some EEG-seizure patterns with short duration, with low amplitude, with circumscribed highly focal activity, with high frequency, and with unusual nonrhythmic morphology as well as EEG seizure patterns obscured by artifact are difficult to detect. This explains why seizures of temporal lobe origin can be detected better than those of extratemporal origin.^{9,27} Specificity (FAR) varies between 0.1/h and 5/h. Low FARs are of critical importance for the acceptance of an algorithm in a clinical setting. Especially if the algorithm is used as an alarm device, high FAR would cause frequent unnecessary responses and actions by EMU personnel during VEM or by caregivers in an outpatient setting. If alarms are brought to the attention of the patient, high FAR would result in unnecessary concerns and anxiety of the patient. Reasons for FAR include normal and abnormal brain activity including sleep activity, rhythmic nonepileptiform activity like frontal intermittent rhythmic delta activity (FIRDA) or temporal intermittent rhythmic delta activity (TIRDA), or various artifacts (including chewing, tooth brushing, repetitive movements, eye movements, and so on).^{9,27} Furthermore, the electrodes used (especially basal temporal electrodes) and the montages applied

(multichannel seizure-detection montage referenced against the average of Fz-Cz-Pz) can influence both sensitivity and specificity.^{7,23,27}

To achieve a stable performance (especially concerning specificities), algorithms need to be tested on a large amount of EEG data comprising the complete VEM period and containing long periods of interictal EEG across all stages of the sleep-wake cycle including any kind of artifacts and nonictal physiological and pathological EEG patterns. Therefore, the algorithm needs to facilitate fast and robust handling of large EEG datasets.^{9,27}

Patient-specific algorithms can even further improve sensitivity and selectivity, but require sometimes complex parameter adjustments and training for individual patients.^{32–34}

Detection algorithms designed as alarm systems need to provide on-line calculation with short detection delays. Reported detection delays of available systems are in order of several seconds.³¹

In conclusion, scalp-EEG–based seizure-detection systems can be clinically useful in the everyday EMU setting, but at the current stage, cannot replace continuous patient supervision and complete visual review of the acquired data by specially trained personnel. In an outpatient setting, application of scalp-EEG–based seizure-detection systems is limited because patients won't tolerate wearing widespread EEG electrode arrays for long periods in everyday life. Recently developed subcutaneous EEG electrodes may offer a solution in this respect.

DISCLOSURE

None of the authors has any conflict of interest to disclose. We confirm that we have read the Journal's position on

issues involved in ethical publication and affirm that this report is consistent with those guidelines.

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